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IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

CrowdSPaFE: A Crowd-sourced Multimodal Recommendation System for Urban Route Safety

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ABSTRACT In the age of information technology, location-based services such as Google Maps, Bing Maps, and Apple Maps have become popular for navigation and traffic information. These services usually consider the shortest path, traffic information, nearby places, and multi-modal alternative route suggestions based on a user's constraints. Nevertheless, these services do not always provide the best choice in terms of "user safety". Recently, some research and mobile applications have considered safety issues. Notably, none of these are capable of adapting to dynamic, conflicting safety features and real-time user feedback. Recently, a population-based algorithm called "SPaFE" has been introduced, which deals with crowd-sourced data along with historic data. This population-based approach, however, does not give more weight to recent feedback than to earlier feedback and does not consider updated crime reports. Furthermore, this approach does not consider distance and performs poorly in the area with insignificant or zero feedback. Considering the above background, we have introduced the population based algorithm "CrowdSPaFE" to adapt with dynamic crime reports, latest feedback, navigation in areas with insignificant feedback, and a trade-off between distance and safety factors. Finally, our empirical results of the "CrowdSPaFE" algorithm depict that it significantly outperforms the state of the art.

INDEX TERMS Crowd-sourcing, Population Based Safe Path Algorithm, Risk Minimization Problem, Ant Colony Optimization, Graph Theory

I. INTRODUCTION

An estimated 55.3 percent of the world's population lived in urban areas, which further estimated to increase to 60 percent in 2030 [1]. Urban areas are expanding progressively because of demographic change and economic growth [2]. However, urbanization brings some drawbacks with it. With the increased number of crimes in the city, safety and security have become the topmost concerns of city dwellers for travelling. Current navigation systems such as Google Maps, Apple Maps, and Bing Maps can not minimize the risk of crimes like theft, pick-pocketing, eve-teasing, snatching, hijack, robbery, unwanted physical touch, rape, murder, etc. To be specific, current widely used navigation systems do not take into account safety factors such as prior crime reports, accidental data, route infrastructure, and other factors associated with a particular path. However, several studies have been conducted to develop a systematic approach to find a safe route considering different safety factors, such as crime data, accidental data, etc. Local government authorities continuously collect and update dataset of accidental data, crime data, and route-related data. Considering these data, several applications have been made to find the route by providing a trade-off between distance and safety [3]–[6].

In reality, crime events depend on multiple factors, such as the user's age, gender, time, road condition, etc. To accommodate the user's personal safety factors, researchers have proposed several algorithms and developed applications [7]. PAROS [8] provides a route relying on the skyline algorithm [9] based on the user's preferences where the user can select criteria among distance, speed, and altitude. PreGo model [10] recommends routes that allow users to add preferences such as time, scenery, and route condition. ComfRide [11] focuses on passengers' personalized comfort factors such as less crowd area, less noise, pleasant roadside scenarios, and less waiting time at stoppages. It utilizes GPS data and inertial sensors to recommend a comfortable route based on a wide range of comfort factors.

With the increasing crime rate, it has become very hard to report, collect, and analyze location-based crime data from a large amount of crime data. Nevertheless, mobile and web application based social media have made it easier to report location-tagged crime incidents as a large number of people can participate in them [12], [13]. Motivated from this, TREADS [14] uses text summarization to extract data

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from Twitter and Yelp, which is then used to recommend a safe route. However, it ignores other safety concerns and only relies on influential Twitter users' tweets within a region. Since all Twitter users do not reside in the same clique, other commuters from different cliques can not reach their information. To be precise, this system often provide incomplete and biased results. SafeJourney [15] and Socroutes [16] utilizes the geo-tagged tweets to make travel more enjoyable and safer. An application [3] has been developed based on social crime reports and geo-coded tweets of Maxico and classifies these data based on the Naive Bayes Classifier. However, these systems can not be used ubiquitously. The Crowdsafe [17] application allows users to report crimes and update crime data while recommending safe routes using the A* and Dijkstra algorithms.

None of the above algorithms take into account multiple conflicting safety factors, have no user data, and do not train on local historical data. Motivated by these limitations "SPaFE" [18] introduced a population-based algorithm based on multi-modal local historical and crowd-sourced data. It codifies multiple conflicting safety factors to construct a single safety value. It assesses a wide range of safety factors and makes a selection from a finite set of options using the TOPSIS algorithm and Entropy Calculation. By modifying the Ant Colony Optimization (ACO) [19] population-based method, it creates a crowdsourced model. Motivated by the behavior of real ant colony, ACO is exploited in artificial colonies and search graphs to find approximate solutions to discrete and continuous optimization problems such as routing and load balancing. It considers the user's feedback as the pheromone of real ants, which further use for searching target. SPaFE generate graph by extracting data from Open Street Map (OSM) API [1] and navigate the safest route in OSM.

SPaFE applied Ant Colony Optimization partially by excluding a significant feature, pheromone evaporation, where pheromone evaporates with time as the latest pheromone has a greater impact on the search path. It does not give priority to recent feedback over older feedback. In addition, it only trained the crime dataset once based on primary local historic data. To be precise, SPaFE does not take into account dynamic crime data or newer crime reports. In SPaFE, feedback can be either positive or negative. It does not consider the degree of positive and negative feedback since feedback can be varied for different experiences of crime and comfort. Moreover, SPaFE does not provide trade-offs between distance and safety.

Motivated by the aforementioned study, we improved the population-based multi-modal algorithm and created an algorithm called "CrowdSPaFE" to include dynamic crime statistics, the most recent feedback, navigation in areas with few people, and trade-offs between safety and distance. To incorporate the most recent user feedback, CrowdSPaFE employs the key component of ACO, pheromone evaporation. Besides in CrowdSPaFE, weighted feedback gives user feedback more importance. CrowdSPaFE classifies crimes according to their severity, which has a big impact on how the risk value is calculated.

Finally, this paper makes the following contributions to the state of the art:

- We modified the generic framework of SPaFE called On- road Risk Minimization Problem (ORMP).
- To solve the ORMP, we provide an algorithm called CrowdSPaFE that takes into account recent feedback, distance, and safety as well as the most recent criminal events and population data.
- Finally, we evaluate CrowdSPaFE's performance, and our empirical results indicate a significant performance improvement over the state-of-the-art.

The rest of this paper is organized as follows. In Section II, we review relevant background and formulate the problem. In Section III, we discuss the details of our proposed algorithm. Next, Section IV presents the empirical evaluation of our algorithm as opposed to the current state-of-the-art. Section V concludes.

II. BACKGROUND AND PROBLEM FORMULATION

In this section, we formulate On road Risk Minimization Problem (ORMP) and review the relevant background to understand our proposed algorithm.

A. ON ROAD RISK MINIMIZATION PROBLEM (ORMP)

The On road Risk Minimization Problem (ORMP) initially formulated in SPaFE. We modify the On road Risk Minimization Problem (ORMP) which defined by the tuple $\langle G, N, E, R, S, F, P \rangle$., Where

- G defines the network of the road.
- N defines a set of nodes of G $\{a_1, a_2, \ldots, a_m\}$ and E defines a set of edges $\{e_1, e_2, \ldots, e_n\}$ where each edges connects two nodes $(a_i, a_j) \in N \mid i \neq j$.
- R is the set of risk values $\{r_{e_1}, r_{e_2}, \ldots, r_{e_n}, r_{a_1}, r_{a_2}, \ldots, r_{a_m}\}$. Each node $a_i \in N$ associated with a risk value r_{a_i} and each edge $e_i \in E$ associated with a risk value r_{e_i}
- S is the set of safety score $\{s_{e_1}, s_{e_2}, \ldots, s_{e_n}, s_{a_1}, s_{a_2}, \ldots, s_{a_m}\}$. Each node $a_i \in N$ associated with a safety score s_{a_i} and each edge $e_i \in E$ associated with a safety score s_{e_i}
- These risk values are computed based on a set of safety factors $F = \{f_1, f_2, ..f_n\}$
- $P_{sd} = \{P_1, P_2, \ldots, P_l\}$ denotes the set of path between source $s \in N$ and destination $d \in N$

The objective of ORMP is to find a path $P_i \in P_{sd}$ such that the corresponding aggregated risk value associated with the path is minimized (Equation 1).

$$P^* = \arg\min_{P_i \in P} \left(\sum_{a_i \in p} r_{a_i} + \sum_{e_i \in p} r_{e_i} \right) \tag{1}$$

A greater safety score and a smaller risk value have a positive impact on the safe route. Risk values are calculated based on the safety factors. It is worth mentioning that it is difficult

[1] https://wiki.openstreetmap.org/wiki/Main$_{Page}$

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to reconcile these contradictory criteria. Taking into account this problem, Multi Attribute Decision Making (MADM) can be modified to make a decision from these conflicting attributes.

B. MULTI ATTRIBUTE DECISION MAKING

In order to comprehensively and objectively reflect safety conditions on roads, it is important to quantify and codify risk values based on safety features F. Therefore, important features are given more priority (or weight) while calculating risk value. The MADM method can be used in this regard. The MADM method assists in finding the best quantitative solution from a set of finite alternatives by necessitating inter-attribute and intra-attribute comparisons. MADM consists of the following parts.

- Features selection and historical data collection
- Decision Matrix with Alternatives and attributes (Criterion).
- Weight calculation of the attributes.
- Measuring the performances of alternatives.

1) Features selection and historical data collection

Road safety depends on several safety factors. Safe route recommendation needs to consider these factors while searching. From this diverse set of safety features, we consider the following set $SF = \{Gender, Age, Time, Severity, Crimetype\}$. The reason behind choosing these safety features is briefly discussed below.

- **Gender:** Men and women may experience different levels of road safety. Women are facing different kinds of harassment, such as eve teasing [20], sexual harassment, abusive attacks [21]. Besides, research shows that road accidents and crashes can vary with gender and age [22], [23]. Here, we set 0 for female and 1 for male.
- **Age:** Due to physical and mental differences, the safety of different ages on the road can vary. Children, young people, and elderly people are more vulnerable [24], [25]. For this reason, several applications [26], [27] have been introduced to increase awareness among children about road safety.
- **Time:** Daytime running lights are strongly related to road safety. Lack of light and poor visibility, which help to increase criminal activity on the road [28]
- **Severity:** The severity of an area or path represents its level of safety. We scale severity one to ten to represent severity in a node or path where ten indicates worse.
- **Crime Type:** People on the road are victims of a wide range of crimes. Some of them are life-threatening, and some of them have less impact. Here, we assign weights to different crimes based on the type of punishment or sentence of imprisonment given to the suspect [6], [29].

The next step is to collect historical crime data from local police stations, newspaper and web scraping. After that, we store these data as alternatives A_i based on above safety features in particular nodes and edges. A risk value is calculated for each alternatives A_i based on the safety features.

2) Decision Matrix with Alternatives and attributes (Criterion).

After collecting historical crime data a decision matrix is constructed as Table **1**.

Alternatives	Attributes			
	$C_1(w_1)$	$C_2(w_2)$	...	$C_M(w_M)$
A_1	m_{11}	m_{12}		m_{1M}
A_2	m_{21}	m_{22}	...	m_{2M}
.	.	.	...	.
A_N	m_{N1}	m_{N2}	...	m_{NM}

TABLE 1: Decision Matrix

Here A_i $(for\ i = 1, 2, ..., N)$ denotes the available alternatives, and C_j $(for\ j = 1, 2, \ldots, M)$ denotes attributes (criteria)on which basis available alternatives are ranked, weights of criteria w_j $(for\ j = 1, 2, \ldots, M)$ and the value of performance of alternatives, DM_{ij} $(for\ i = 1, 2, \ldots, N; j = 1, 2, \ldots, M)$. The decision matrix is used for finding the best alternative and ranking them. However, risk factors can change over time. Risk can be minimized through effective law enforcement, improved road infrastructure, and transportation systems. Recent events have more impact than earlier events. To be precise, latest events contributes more to safe path algorithm. A set of incidents $in = \{in_1, in_2, \ldots, in_k\}$ associated with each nodes and edges. Each incidents $in_i \in in$ consists of set of safety factors. Now calculate weighted average of each associated feature's $f_i \in F$ using the Equation 2

$$\overline{f_i} = \frac{f_i * (1+\gamma)^t + f_i * (1+\gamma)^{t-1} + \ldots + f_i * (1+\gamma)^1}{t} \tag{2}$$

Here, γ is growth factor. After calculating the weighted average, we create new decision matrix DM' is constructed. It is worth mentioning that we normalized the elements of all decision matrices to produce the same unit for the attributes. The next two steps of MADM consist of weight calculation and measuring the performance of alternatives that are calculated from these decision matrices.

3) Weight calculation of attributes

In MADM, weight assignment is one of the toughest jobs regarding which alternatives are to be ranked. There are two types of methods for determining the weight of the attributes. These are subjective and objective. Subjective fixed-weight methods such as the analytic hierarchy process method (AHP) [30], the Delphi method [31], expert survey method etc. On the other hand, objective fixed weight methods are based on the inherent information of attributes to calculate

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the weights of attributes, which helps to remove man-made disruption and make results more yielding to the facts. On that account, the objective fixed weight method "Entropy Method" is used in this research. The steps in the "Entropy Method" are as follows:

1) **Standardization:** At first scale the measured values [0-1] of attributes. Calculate the feature weight p_{ij} for the jth attributes of ith alternative as follow:

$$p_{ij} = \frac{m_{ij}}{\sum_{i=1}^{n} m_{ij}^2} \tag{3}$$

2) **Entropy Calculation:** Next step is the calculation of the Entropy, e_j of the ith attribute in p_{ij}:

$$e_j = -k \sum_{i}^{N} (p_{ij}\ ln\ p_{ij}) \quad , k = \frac{1}{ln_N} \tag{4}$$

where k guarantees $0 \leq e_j \leq 1$. The divergence of coefficient of jth (d_j) calculated as follow:

$$d_j = |1 - e_j| \tag{5}$$

3) **Weight Calculation:** The objective weight for each attribute $C_j(for\ j = 1, 2, 3, \ldots, M)$ calculated as follow:

$$w_j = \frac{d_j}{\sum_{k=1}^{M} d_k} \tag{6}$$

4) Measuring the performances of alternatives

TOPSIS method is often used to solve MADM problems by finding the best option from a set of alternatives and ranking them. Here we tailored TOPSIS to calculate the risk values based on the safety features. We applied TOPSIS on weighted decsion matrix $DM^{'}$. TOPSIS method consists of the following steps.

1) **Normalize Decision Matrix:** First step is to normalize decision matrix based on following equation.

$$m_{ij}^{*} = \frac{m_{ij}}{\sqrt{\sum_{1}^{m} {m_{ij}}^2}} j = 1, 2, ...n \tag{7}$$

2) **weighted normalized matrix:** Now, construct weighted normalized matrix X_{ij} by multiplying weight with value of attributes following the below equation.

$$X_{ij} = w_j . m_{ij}^{*} \tag{8}$$

3) **Calculate worst and best alternative:** Determine the worst $A_w$alternatives and best A_b alternatives from the newly constructed weighted matrix based on the following equation.
A_w = { $\langle max(\mathrm{X}_{ij}|\mathrm{i}= 1,2,3, \ldots ,\mathrm{m})|\mathrm{j} \in J_{-}\rangle, \langle min(\mathrm{X}_{ij}$ |i= 1,2,3,.. ,m) |j $\in \mathrm{J}_{+}\ \rangle\} \equiv \{\mathrm{X}_{wj}|\mathrm{j}=1,2,..,\mathrm{n}\}$,

A_b = { $\langle min(\mathrm{X}_{ij}|\mathrm{i}= 1,2,3, \ldots ,\mathrm{m})|\mathrm{j} \in J_{-}\rangle, \langle max(\mathrm{X}_{ij}|\mathrm{i}=$ 1,2,3, .. ,m)|j $\in \mathrm{J}_{+}\ \rangle\} \equiv \{\mathrm{X}_{bj}|\mathrm{j}=1,2,..,\mathrm{n}\}$
where,

$J_{+} = \{j = i = 1, 2, .., n|j\}$ associated with positive impact of attributes, and

$J_{-} = \{j = i = 1, 2, .., n|j\}$ associated with negative impact of attributes.
In our work, J_{+} will be an empty set, $J_{+} = \emptyset$ and J_{-} will contain the all safety features as we want to minimize each feature in the safety set.

4) **Calculate distance from worst and best alternative:** To codify the risk value, eucledian distance from the ideal best A_b or ideal worst A_w for each alternatives.

$$d_{iw} = \sqrt{\sum_{n}^{j=1} (X_{ij} - X_{wj})^2}, i = 1, 2, .., m \tag{9}$$

$$d_{ib} = \sqrt{\sum_{n}^{j=1} (X_{ij} - X_{bj})^2}, i = 1, 2, .., m \tag{10}$$

5) **calculate the relative closeness:** For each alternatives calculate the relative closeness from the ideal solution denoted as R_i.

$$R_i = \frac{d_{iw}}{d_{iw} + d_{ib}} \tag{11}$$

Table 2 contains the risk values for each alternative.

Alternatives	Risk Values
A_1	R_1
A_2	R_2
.	.
A_N	R_n

TABLE 2: Safety Scores of Alternatives

C. POPULATION BASED ALGORITHM

In a population-based algorithm, each individual contributes to the problem in order to achieve the optimal solution. Contributions from the population help to move towards a better solution gradually. One of the population-based algorithms is the Ant Colony Optimization Problem (ACO). Ant Colony Optimization Problem inspired from the foraging behaviour of ants. Ants leave pheromones on the ground in order to mark the trails, which enables them to communicate with other ants and helps them to find the shortest path from the nest to food sources. The main characteristic of ACO problems is that at each iteration, the pheromone intensity of the trail is updated by pheromone evaporation and builds a solution in the solution itself. Pheromone evaporation is a main feature of ACO, where pheromone evaporates with time. Therefore, the older pheromone loses importance with time. On the contrary, the importance of the latest pheromone increases. It evaporates pheromone density as Equation 12

$$\tau_{ij} \leftarrow (1 - \rho).\tau_{ij} + \sum_{k=1}^{m} \Delta\tau_{ij}^{k} \tag{12}$$

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Here, τ_{ij} is current pheromone density associate with edge that links node i and j and m is number of ants, $\Delta\tau_{ij}^{k}$ is pheromone on edge(i,j) laid by k^{th} ant and ρ is the evaporation rate. An improved version of ACO is *MAX–MIN Ant System (MMAS)* where pheromone update is bound to lower limit and upper limit which follows Equation 13

$$\tau_{ij} \leftarrow \left[(1-\rho).\tau_{ij} + \sum_{k=1}^{m} \Delta\tau_{ij}^{k}\right]_{\tau_{min}}^{\tau_{max}} \tag{13}$$

where τ_{min}, τ_{max} lower and upper bounds imposed on total pheromone.

In the context of safe road recommendations, we can use the idea of ACO where each commuter contributes to the solutions. Local historical data will not be enough of a safe road recommendation as many crime incidents are not reported to police stations due to various reasons, such as complex system of crime reporting, time-consuming, political influence, fear etc. Besides, local newspapers and social media highlight big crimes where small crimes are not published. Therefore, local users and commuters can play a big role in this context. After the safe route recommendation, the user's feedback can be counted as the pheromone of the ants. As users' recent feedback has more impact than earlier ones, we use pheromone evaporation of ACO. In our research, we utilize ACO and modify the Equation 13 as Equation 14

$$Sf_i \leftarrow \left[(1-\rho)Sf_i + \delta * \sum_{k=1}^{\beta} \Delta Sf_i^{k}\right]_{Sf_{min}}^{Sf_{max}} \tag{14}$$

In this approach safety score updates on user's feedback. Here we use evaporation method of ACO to mitigate impact of older feedback and increase impact of latest feedback after β number of feedback. Here Sf denotes the safety feedback.

D. SPAFE ALGORITHM

SPaFE proposed a population-based algorithm which applies multiple attribute decision making (MADM) to select features from multiple conflicting safety features and weights them accordingly to their impacts. MADM applied to select a process from a limited number of alternatives. This method uses historical crime data from law enforcement, newspapers, and other crime sources to compute risk values at each node and edge of the route. In addition, it takes into account the user's feedback on the route by adopting a population-based approach called Ant Colony Optimization (ACO) and updates the risk value. An overview of the SPaFE algorithm is shown in Algorithm 3

In the SPaFE algorithm, first collect historical data and then construct decision matrices DM. Calculate risk value using MADM as discussed earlier, which is shown in Algorithm 2. Then for each query it applied A* search based on risk score of stored in the nodes and edges of the graph. Based on A* search, it returns a path for a fixed source and destination with a minimum aggregated risk value. Afterward,

Algorithm 1: SPaFE (Source s, Destination d)

Data: Historic Data, DM
Result: P_{sd}

```
1  R ← MADM(DM)
2  PositiveFeedbackCount ← 0
3  NegativeFeedbackCount ← 0
4  TotalQueryCount ← 0
5  while User's Query Continue do
6      P_st ←A*(R);
7      if positive feedback then
8          update for every node a_i ∈ P_sd and edge e_i
             ∈ P_sd
9          PositiveFeedbackCount ←PositiveFeedbackCount+
             1
10     end
11     if negative feedback then
12         update for every node a_i ∈ P_sd and edge e_i
             ∈ P_sd
13         NegativeFeedbackCount ←
             NegativeFeedbackCount + 1
14     end
15     TotalQueryCount =
         PositiveFeedbackCount +
         NegativeFeedbackCount
16     if QueryCount ≥ β then
17         for each node and edge in G
18           r_i =
             r_i − δ ∗ (getPositiveFeedbackCount(i) −
             getNegativeFeedbackCount(i))
19     end
20 end
```

it takes feedback from the user based on their experience. Now, it updates the feedback count for each node and edge of the safe path P_st. If feedback is positive, it increments the value of the positive feedback count. On the other hand, for negative feedback, it increment the negative feedback count. If total user's feedback is greater than the defined threshold β, then it updates the risk values r_i for all nodes $a_i \in G$ and all edges $e_i \in G$ based on the previously recorded user's responses in $PositiveFeedbackCount$ and $NegativeFeedbackCount$ which shown in Line 18. Here, δ determine the degree of change of the risk value R of a node a_i or edge e_i based on the user's feedback. SPaFE algorithm return safest path for source s and destination d based on query $q = \{s, d\}$. Precisely, with the user's feedback SPaFE calculated a new safe path based on new update risk value.

III. THE CROWDSPAFE ALGORITHM

SPaFE algorithm solved various safety concerns in the safe route recommendation of previous works. However, SPaFE algorithm has some limitations. It does not consider dynamic crime data or accidental data from law enforcement, newspaper and other sources. The impact of recent crimes or accidental events is greater than earlier

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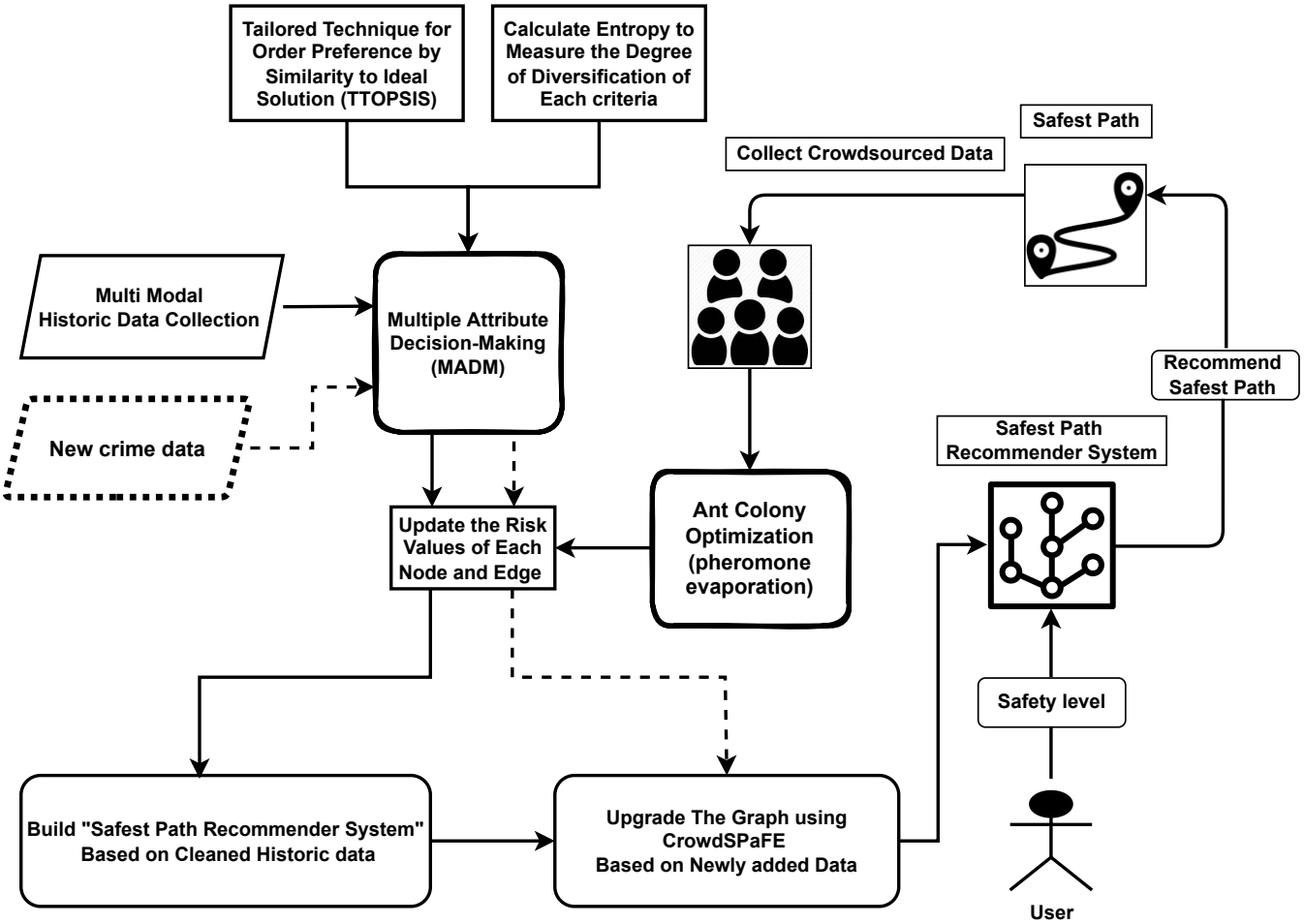


FIGURE 1: Overview of "CrowdSPaFE".

Algorithm 2: MADM (Decision Matrix DM)

1 **Weighted Data,** $DM^{'}$ $\leftarrow WeightedAverageofSafetyFeatures$(DM)
2 $w_i \leftarrow Entropy$(DM$^{'}$);
3 **R** $\leftarrow TOPSIS$(DM$^{'}$, w_i)
4 return **R**

incidents on the possibility of new crimes or accidental data. To overcome these limitations, we propose an algorithm called "CrowdSPaFE" that dynamically updates the risk value of each node and edge of the route network after a number of new reported crimes and accidents and weights them considering their time of incident. As a result, it mitigates the impact of older crime data and intensifies the impact of the latest crime data. Furthermore, SPaFE partially implements ACO where it considers user feedback as pheromone in ACO and updates the risk value with $\delta * (positiveFeedback - negativeFeedback)$. Pheromone evaporation is a main feature of ACO, where pheromone evaporates with time. Therefore, the older pheromone loses importance with time, while the importance of the latest pheromone increases. Following the pheromone evaporation equation of ACO CrowdSPaFE, it updates its risk value as Equation 14. Moreover, CrowdSPaFE accepts weighted feedback from users who can rate the comfort, discomfort, and severity of crime or accident (-5 to 5) with the value -5 denoting the worst experiences with crime events such as rape, sex crime, murder, etc. and the value 5 denoting the best experience, which indicates the best route infrastructure, better transportation, better security, etc. The SPaFE algorithm, on the other hand, only accepts unit feedback, which is either a 1 or a -1 for any kind of user experience.

The algorithm for CrowdSPaFE is shown in Algorithm 26. At the beginning of the algorithm, it collects local historical data and constructs decision matrices. In line 2, it calculates the risk value of the alternatives based on MADM, which is shown in Algorithm 2. Algorithm 2 takes a decision matrix DM and calculate the risk value based on MADM discussed in II-B. Line 3-5 it set safety feedback ΔSf of each nodes and edges to 0 and set total query count and newly reported crime events count to 0. ΔSf stores change of safety feedback since the last update of safety feedback

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Algorithm 3: CrowdSPaFE (Source s, Destination d, safety level σ)

Data: Historic Data, DM
Result: P_{sd}

1 $R \leftarrow MADM(\mathrm{DM})$
2 $\Delta Sf \leftarrow 0$
3 $QueryCount \leftarrow 0$
4 $NewReportedCrimeCount \leftarrow 0$
5 **while** *User's Query Continue* **do**
6 $\quad \mathrm{P}_{st} \leftarrow \mathrm{A}^*(G, S, \sigma)$;
7 $\quad$ **if** *user gives positive weighted feedback* f_+ **then**
8 $\quad\quad$ update for every node $a_i \in P_{sd}$ *and edge* $\mathrm{e}_i \in P_{sd}$
9 $\quad\quad\quad \Delta Sf_i = \Delta Sf_i + f_+$
10 $\quad$ **end**
11 $\quad$ **if** *user gives negative weighted feedback* f_- **then**
12 $\quad\quad$ update for every node $a_i \in P_{sd}$ *and edge* $\mathrm{e}_i \in P_{sd}$
13 $\quad\quad\quad \Delta Sf_i = \Delta Sf_i - f_-$
14 $\quad$ **end**
15 $\quad$ **if** *new crime reported* **then**
16 $\quad\quad$ update Decision matrix, DM
17 $\quad\quad NewReportedCrimeCount = NewReportedCrimeCount + 1$
18 $\quad$ **end**
19 $\quad QueryCount = QueryCount + 1$
20 $\quad$ **if** $QueryCount \geq \beta$ *or* $newReportedCrimeCount \geq \gamma$ **then**
21 $\quad\quad R^{'} \leftarrow MADM(\mathrm{DM}^{'})$
22 $\quad\quad$ update for every node $a_i \in P_{sd}$ *and edge* $\mathrm{e}_i \in P_{sd}$
23 $\quad\quad\quad Sf_i = [(1-\rho)Sf_i + \delta * \Delta Sf_i]_{Sf_{min}}^{Sf_{max}}$
24 $\quad\quad\quad S_i = r_i - Sf_i$
25 $\quad$ **end**
26 **end**

score Sf. Lines 5–20 are executed for every query by users. When a user queries for a safe route between a source and a destination in line 6, it applied A* search to find a path P_{sd} with safety level σ shown in line 7. Here σ used for trade-offs between distance and safety. The objective function of A* in CrowdSPaFE shown in Equation 15.

$$\underset{s \rightarrow t}{MIN} \left\{ \sum_{e \epsilon E} (1-\sigma).length(e) + \sigma.risk(e) \right\} \tag{15}$$

The safety factor, which ranges from [0,1], determines the desired level of safety for users. The desired safety level can vary from user to user. Some users' request routes with the shortest routes also minimize risk. On the other hand, some users may request route recommendations with the highest safety without worrying about distance. CrowdSPaFE enables users to find routes based on their preferences. On the contrary, A* search in SPaFE provides the route with the lowest risk value without taking distance into account. The objective function of this A* search shown in Equation 16

$$\underset{s \rightarrow t}{MIN} \left\{ \sum_{e \epsilon E} risk(e) \right\} \tag{16}$$

After traveling through the recommended route, users can give feedback about the route. Here, we consider user feedback as the pheromone of ACO. Users provide feedback based on their experiences, which vary from -5 to 5. In line 7-13, it updates the change of safety feedback ΔSf for every nodes and edges of path P_{sd}. For positive weighted feedback, it increases the value of safety feedback ΔSf by feedback weight and decrease the safety feedback for the negative weighted feedback. In lines 15–18, if new crime events or accidents occur, then it updates the decision matrix of that particular node or edges where the incident occurred. Newly reported incident has more impact on the safety of the route. This newly created decision matrix is further used to calculate a new risk value using MADM. As a result, the impact of previous crime events on risk value diminishes, while the impact of recent crime events on risk value increases. In line 20, it checks whether the user's total responses exceed the defined threshold β, or new crime reports exceed defined threshold γ. In line 21, it recalculates the risk value using the MADM approach for new decision matrices. In line 23, it updates the safety feedback value of Sf_i for all nodes and edges of G based on Equation 14. Here we bound safety feedback value within Sf_{max} and Sf_min because the safety feedback value Sf_i may become a outlier for areas with large amount of users than areas with smaller number of users. This may happen because large users update the safety feedback value Sf_i more. As a result, A* search will be biased to these outlier values. Thus, users' in small user area or zero user areas may benefit less. To be precise, we bound the safety feedback value to $[Sf_{max}, Sf_{min}]$ to control the safety feedback value. Now, in line 24, it updates the safety score for all nodes $a_i \in G$ and edges $e_i \in G$ by subtracting feedback value from the risk value in line 23. Positive safety feedback minimizes the risk value, while negative feedback increases the risk value. The value δ is a predefined value to determine the degree of change of the safety feedback value R of a node a_i or edge e_i based on user's feedback. This safety score S further used in A* search in new query.

The CrowdSPaFE algorithm is divided into two parts: risk value updates and path finding. Risk value calculated based on MADM which consist of Entropy method and TOPSIS. Time and space complexity of both method is $O(n*m)$ where n rows represent node or edge and m columns represent attribute of corresponding node or edge. Complexity for update safety feedback score is $O(|N| + |E|)$ where N is set of nodes and E is the set of edges. Time complexity for querying safest path once for a single source-destination pair is $O(|E| * log(|N|))$ for graph with N nodes and E edges.

In summary, each updates incorporates new crimes, accidental data and user's weighted feedback based on their travel

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This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2023.3252881

experience helps CrowdSPaFE algorithm to calculate new safe path. Overview of the CrowdSPaFE approach shown in Figure 1.

IV. EMPIRICAL EVALUATION

In this section, we empirically evaluate and analyze the performance of our proposed algorithm CrowdSPaFE and compare it with the performance of the SPaFE algorithm in terms of safety. In our experiment, we considered two types of graphs. The first graph is a small complete graph, which consists of 1000 nodes where each node has 2-4 edges. The second graph is a large graph which consists of 5000 nodes where each node has 2-4 edges. For various experimental setups, we evaluate the path change of the CrowdSPaFE from the SPaFE path. We use Python to implement our approach and to conduct all experiments on a machine with a Intel Core i7-8700 CPU , 32GB RAM and Mesa Intel UHD Graphics 630 (CFL GT2).

A. EXPERIMENT E1: COMPARE CROWDSPAFE WITH SPAFE FOR USER'S RESPONSE

In Figure 2, we evaluate the change in the path of CrowdSPaFE from the path of SPaFE with the change of the user's feedback. In Figure 2(a) we calculate path change for the small graph, and in Figure 2(b) we calculate path change for the large group. In SPaFE risk value updates with user feedback following the equation $r_i = r_i - \delta * (positiveFeedbackCount(i) - negativeFeedbackCount(i))$ which does not consider the latest feedback and feedback of users can be either 1 or -1. On the other hand, CrowdSPaFE updates risk values with the user feedback as Equaiton 13 which results in emphasizing the latest feedback and mitigating the impact of earlier feedback. Besides, it enables users to give weighted feedback of -5 to 5 based on their experience. As a result, risk value updates for CrowdSPaFE are more significant than for SPaFE. In both graphs, we evaluate the total path change after 100, 200, 300 , 400, and 500 feedbacks, respectively. According to experimental results, we can report a 20–65% SPaPE path change for CrowdSPaFE in the small graph and a 10–50% SPaPE path change for CrowdSPaFE in the large graph. The risk value of the small graph is updated more frequently than the large graph. As a result, the small graph path change rate is higher.

B. EXPERIMENT E2: COMPARE CROWDSPAFE WITH SPAFE FOR NEW CRIME EVENTS REPORT

In Figure 3, we evaluate the change in the path of CrowdSPaFE from the path of SPaFE with newly reported crime and accident incidents . In Figure 3(a) we calculate the change of path for the small graph and in Figure 3(b) we calculate the change of path for the large group. In SPaFE risk value is calculated once for historical crime data at the beginning of the algorithm. On the other hand, CrowdSPaFE support dynamic crime events and accidental data updates. Since CrowdSPaFE calculates risk value using MADM after a certain number of crime events reported or accidents reported, it mitigates the impact of older incidents and focuses more on the latest incidents, which will be more significant in safe path recommendation. In both graphs, we assess total path change after 100, 200, 300, 400, and 500 reported crime events, respectively. Here, we add crime events to either the nodes or edges randomly. Based on experiment results, we can report a 10–60% SPaPE path change for CrowdSPaFE in the small graph and a 5-40% SPaPE path change for CrowdSPaFE in the large graph. The risk value of the small graph is updated more frequently than the large graph. As a result, the small graph path change rate is higher the large graph.

C. EXPERIMENT E3: COMPARE TOTAL AGGREGATED DISTANCE OF CROWDSPAFE PATH WITH SPAFE FOR DIFFERENT SAFETY FACTOR σ.

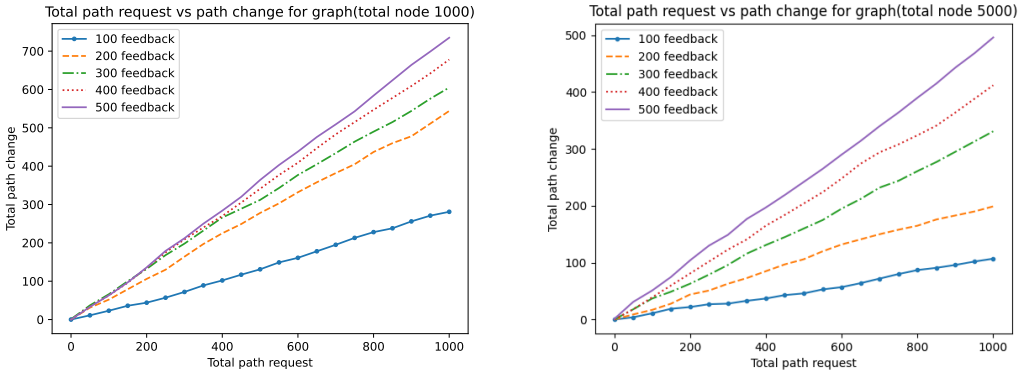
In Figure 4, we evaluate the total aggregated distance of CrowdSPaFE with varying safety scores and compare it with the SPaFE path. Both the CrowdSPaFE and the SPaFE approaches use A* search to find a safe path. CrowdSPaFE enables users to find the safe route by a trade-off between distance and risk value by adjusting the safety factor σ. We evaluate the CrowdSPaFE approach with different safety factors of 0.2, 0.5, and 0.8 in Figures 4(a), (b), and (c). Safety factor σ is used to reduce the distance of the safe path by compromising safety. From these figures we can visualize that the aggregate distance of the CrowdSPaFE path much smaller than the aggregate distance of the SPaFE. In Figures 4 (a), (b), and (c), we calculate the path for the large group and aggregate the distance and compare it with distance of the SPaFE path. If the safety factor σ is equal to 1 then it provides same route for both approach. In this experiments we consider 0 user feedback and 0 new crime event report since they change the risk value of the graph.

V. CONCLUSIONS AND FUTURE WORK

This paper developes CrowdSPaFE algorithm for recommending safe routes that is crowd-sourced, adaptive and dynamic. We took a comprehensive safety concern and included various conflicting safety factors in this approach. We applied multi-attribute decision making to quantify and codify risk values based on safety features and rank the crime incidents. Furthermore, we adopt the popular population-based algorithm Ant Colony Optimization (ACO) to incorporate users' responses to safe route suggestions and update the risk value of the road network. It also enables users to find a path based on their preferences. From the empirical results, we can conclude that by using CrowdSPaFE, we can notably reduce on road risks and helps city-dwellers navigate safely in the urban environment. Future extensions of our current methodology will incorporate real historical data as well as a knowledge extraction component to collect data on natural feedback of real user and incorporate CrowdSPaFE in Open Street Map.

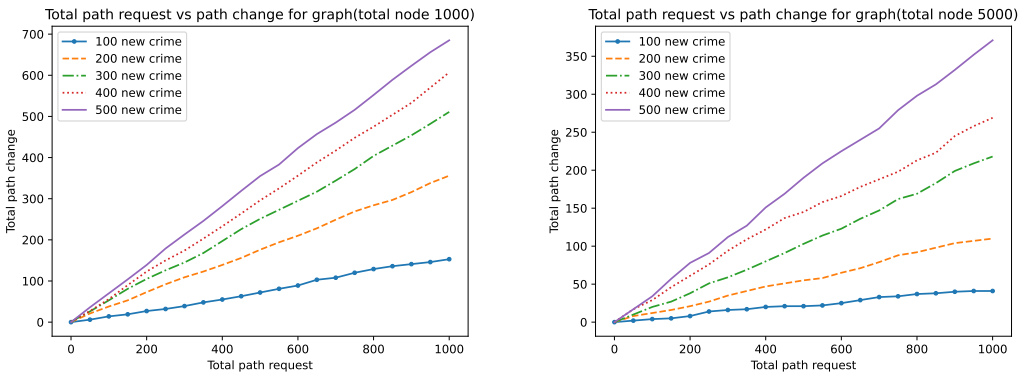
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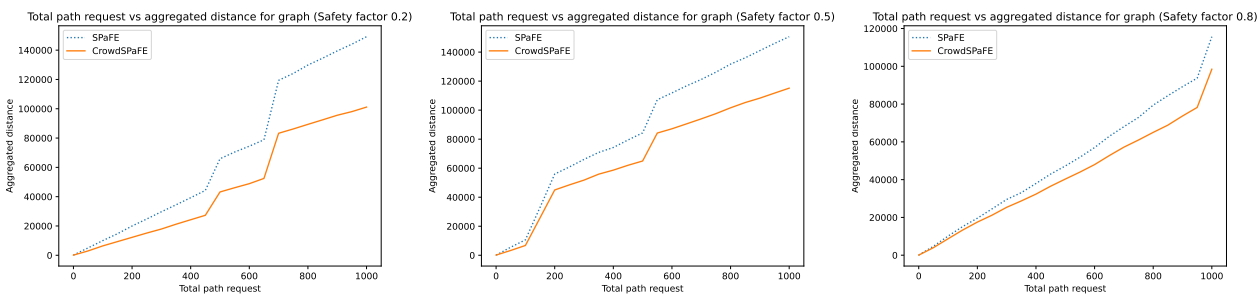
(a) (b)

FIGURE 2: Path updates for CrowdSPaFE after user's feedback. (a) For graph with 1000 nodes. (b) For graph with 5000 nodes.



(a) (b)

FIGURE 3: Path updates CrowdSPaFE after new crime events. (a) For graph with 1000 nodes. (b) For graph with 5000 nodes



(a) (b) (c)

FIGURE 4: Total aggregated distance of CrowdSPaFE path vs Total distance of SPaFE path. (a) Safety factor 0.2. (b) Safety factor 0.5. (c) Safety factor 0.8.

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